Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

P3 — Equations and Information Tables

Equations three to use in P3

$$\rho = \frac{m}{V}$$

density = mass ÷ volume

ρ *in* kg/m^3, m *in* kg, V *in* m^3

Recall

$$\Delta E = m\,c\,\Delta\theta$$

change in thermal energy = mass × specific heat capacity × temperature change

ΔE *in* J, c *in* $J/kg\,°C$, $\Delta\theta$ *in* °C

On the equation sheet

$$E = m\,L$$

thermal energy for a change of state = mass × specific latent heat

E *in* J, m *in* kg, L *in* J/kg

On the equation sheet

Units convert first, then substitute

grams to kilograms	kilojoules to joules	cm^3 to m^3	temperature change
÷ 1000	× 1000	÷ 1 000 000	final − initial

Specific heat capacity, c energy to raise 1 kg by 1 °C

Material	c in J/kg °C
Water	4200
Aluminium	900
Concrete	800
Iron	450
Copper	385
Lead	130

Specific latent heat, L energy to change the state of 1 kg with no change in temperature

Substance	Fusion (solid ↔ liquid) in J/kg	Vaporisation (liquid ↔ gas) in J/kg
Water	334 000	2 260 000
Ethanol	108 000	840 000
Lead	23 000	870 000
Aluminium	400 000	10 500 000

Which equation? Temperature changes: use $\Delta E = m\,c\,\Delta\theta$ (the sloping parts of a heating graph). The state changes at constant temperature: use $E = m\,L$ (the flat parts). Several stages: one equation per stage, then add.